*(d) What is the *peak value* of the sinusoidal current flowing through the heater when switch S is closed ? (2 marks)

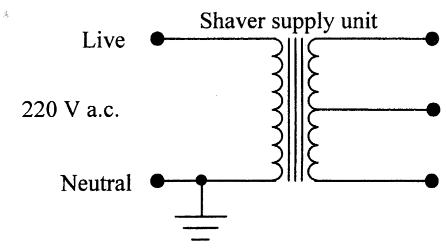
Answers written in the margins will not be marked.

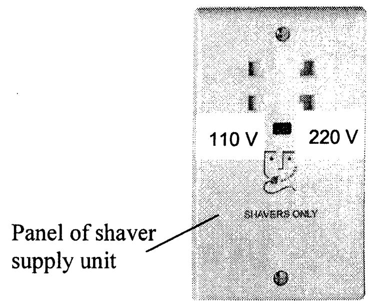
9. Read the following description about the 'shaver supply unit' in bathrooms and answer the questions that follow.

The danger of electric shock is particularly high in bathrooms. Normal electric socket outlets should not be installed in bathrooms. As electric shavers and toothbrushes are becoming popular these days, a special unit, called 'shaver supply unit' is now common in bathrooms to provide electricity just for these low power consumption electric appliances (Figure 9.1).

The shaver supply unit consists of a transformer in which the secondary is not earthed and is completely isolated from the 220 V a.c. mains supply connecting to the primary. It can be used with 220 V or 110 V shavers.

Figure 9.1





(a) Explain why the chance of electric shock is high in bathrooms. (2 marks)

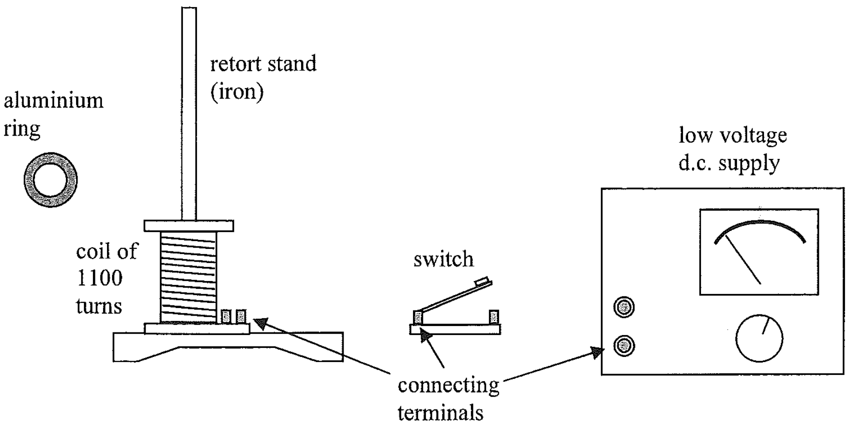
Answers written in the margins will not be marked.

(d) (i) If fuses 3 A, 5 A and 13 A are available, determine which will be the most suitable to limit an excess current. Show your work. (3 marks)

(ii) A student claims that since a.c. is used for the heater, the switch S can be installed in either wire A or wire B. Comment on this claim. (2 marks)

(iii) If a fault resulted in the live wire having contact with the metal case of the heater, which wire, A, B or C, could prevent an electric shock of a person touched the case of the heater ? Explain. (2 marks)

9. (a) You are given a low voltage d.c. supply, an aluminum ring, a switch, a coil of 1100 turns and a retort stand arranged as shown. Use three connecting leads to complete the connections among the apparatus in the figure and describe how to demonstrate Lenz's law in electromagnetic induction. State and explain the observation. (6 marks)



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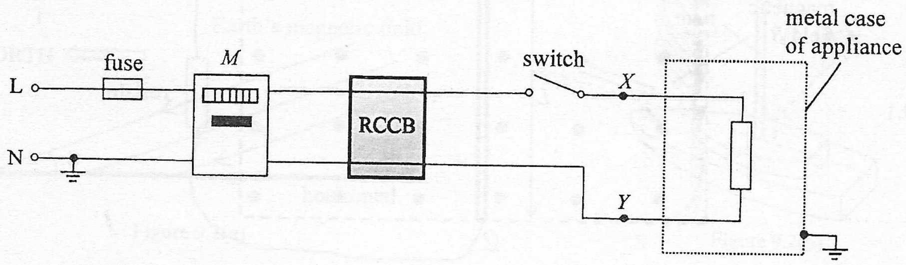
(b) Figure 8.2 shows a simplified domestic circuit connected to an electrical appliance via a fuse, a meter M, a residual current circuit breaker (RCCB) and a switch.

Figure 8.2



(i) What physical quantity does the meter M record ? (1 mark)

....

....

(ii) An RCCB is a kind of safety device that cuts off the supply automatically whenever there is a small difference between the currents in the live (L) and neutral (N) wires. State, in each of the following situations, which device(s) will respond (i.e. the fuse blows and/or the RCCB cuts off the supply).

(1) A short circuit occurs between points X and Y. (1 mark)

....

....

(2) A short circuit occurs between point Y and the metal case of the appliance. (1 mark)

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9. (a)

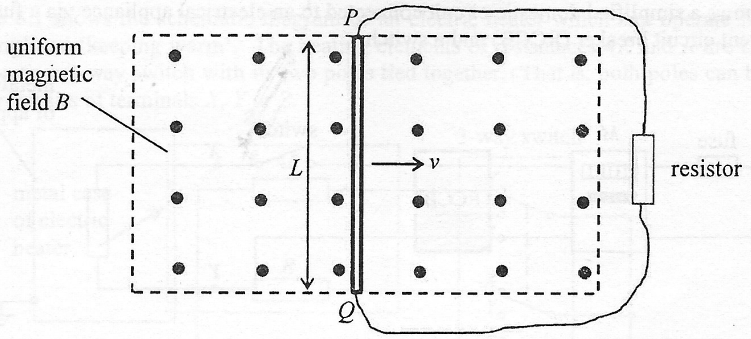

Figure 9.1

Figure 9.1 shows a metal rod PQ of length L moving with constant velocity v across a uniform magnetic field B pointing out of the paper. An e.m.f. ξ is induced across rod PQ as it cuts the field lines. When the rod is connected to a resistor outside the field, a current I flows in the circuit.

(i) Indicate the direction of I in Figure 9.1. (1 mark)

(ii) Explain why an external force F is required to maintain the uniform motion of rod PQ. Find F in terms of the physical quantities given. (3 marks)

(iii) This set-up works as a generator. By considering the mechanical power input by external force F to the set-up, show that $\xi = BLv$. (2 marks)

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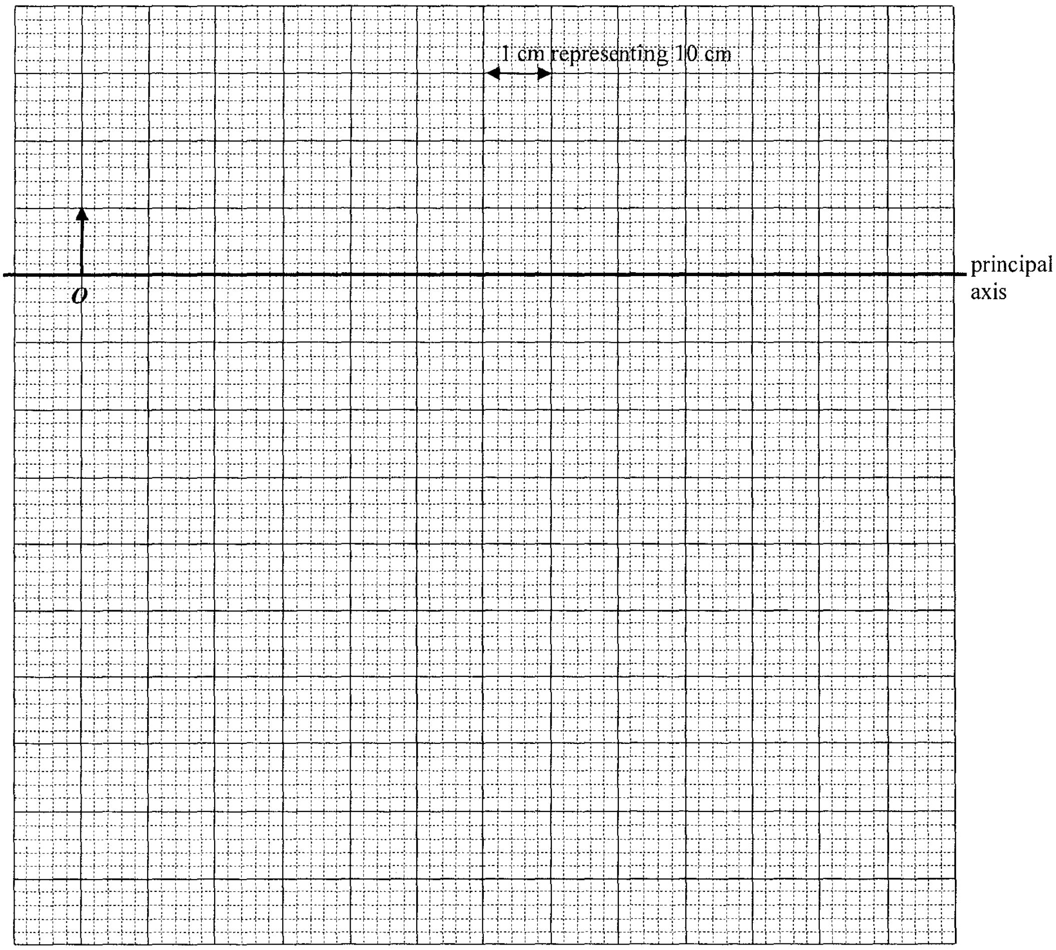
(iii) Draw a ray diagram to show how the image of slide O is formed on the screen. Locate the focus F of lens L on your diagram and find the focal length of L. (Slide O and the principal axis of the lens have been drawn for you.) (5 marks)



Focal length of L = ...

(iv) When a black-and-white slide is projected onto the screen, the image has colour edges. Explain briefly. (Hint: the lens is made of glass.) (2 marks)

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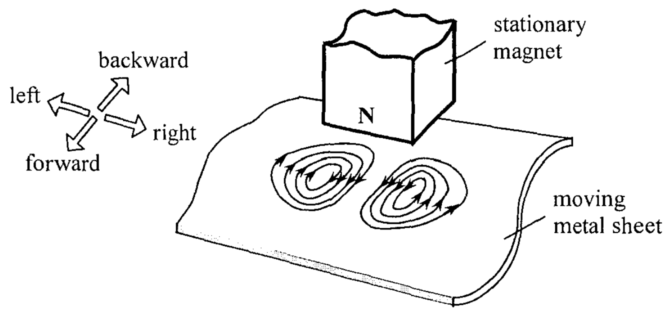
7. Read the following passage about **eddy currents** and answer the questions that follow.

> Eddy currents are induced by changing magnetic fields. They flow in closed loops in conductors like swirling eddies in a stream, perpendicular to the direction of the magnetic field. They are commonly applied in braking known as 'eddy braking'.
>
> The heating effect of eddy currents is used in induction heating devices, such as induction cookers. The resistance felt by the eddy currents in a conductor causes Joule heating. However, for applications like motors and transformers, this heat is considered as a waste of energy and as such, eddy currents need to be minimized.
>
> Eddy currents can be removed by cracks or slits in the conductor, which prevent the current loops from circulating. This means that eddy currents can be used in detecting defects in materials. The magnetic field produced by the eddy currents is measured, where a change in the field reveals the presence of an irregularity in the material.

(a) (i) In Figure 7.1, a permanent magnet with north pole facing downwards is held stationary. A metal sheet moves past the magnet (the direction of movement is not shown) and eddy currents are induced as shown. Briefly explain why eddy currents are induced and state whether the metal sheet is moving forward, backward, towards the left or towards the right. (2 marks)

Figure 7.1



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(ii) State the energy changes in the process in which the metal sheet is slowing down to stop. (2 marks)

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5.

Figure 5.1

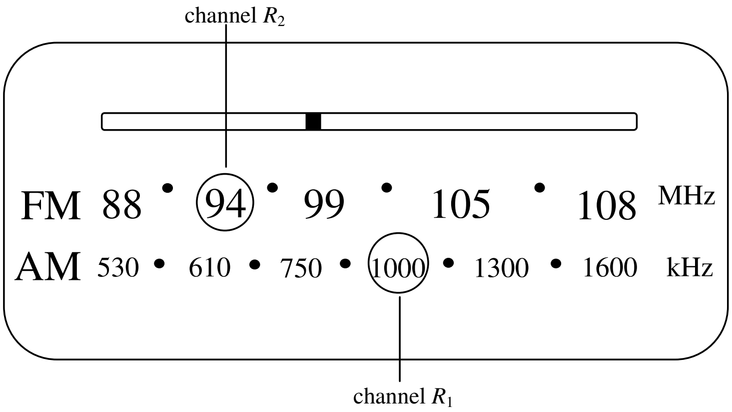

Figure 5.1 shows the display panel of a radio and the broadcasting frequencies of two radio channels R_1 and R_2. Given : speed of electromagnetic waves = 3.00×10^8 m s^{-1}

(a) Find the wavelength of the radio waves used by channel R_1. (1 mark)

(b) Anita's house is surrounded by hills and at her house, the reception of one of the two radio channels is better. For which radio channel is the reception better ? Explain your answer. (2 marks)

6.

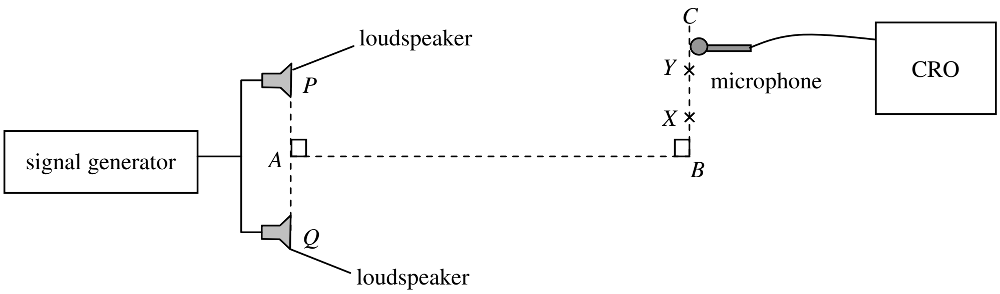

Figure 6.1

Figure 6.1 shows two identical loudspeakers *P* and *Q* are connected to a signal generator. Position *A* is the mid-point of *PQ*. A microphone connected to a CRO is moved along *BC*. The amplitude of the CRO trace increases as the loudness of the sound detected increases. Figure 6.2 shows how the amplitude of the CRO trace varies with the position of the microphone.

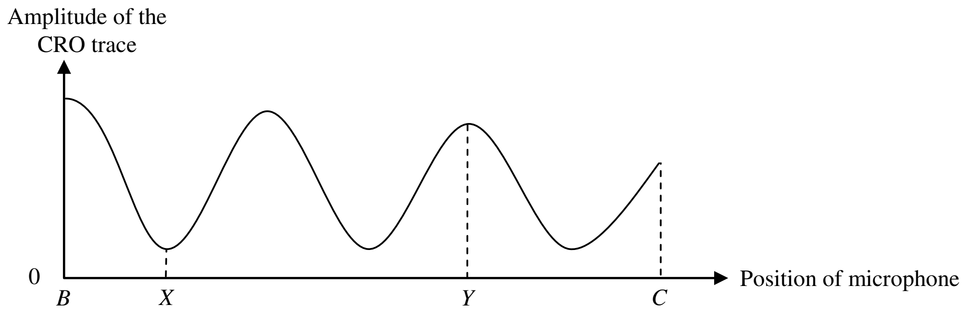

Figure 6.2

(a) (i) Explain why the loudness of the sound varies along *BC*. (2 marks)

....

....

....

....

(ii) State **ONE** reason why the amplitude of the CRO trace is **NOT** zero at position *X*. (1 mark)

....

....

(b) If $PY = 5.10$ m and $QY = 5.78$ m, find the wavelength of the sound. (2 marks)

....

....

....

(b) The spacecraft of mass 7.80×10^3 kg now enters a circular orbit of radius r around the Earth.

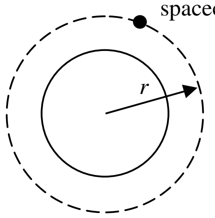

Figure 11.2

*(i) Show that the speed of the spacecraft in the orbit is given by $\sqrt{\frac{g}{r}}R_E$ where R_E is the radius of the Earth. (2 marks)

*(ii) How long does it take for the spacecraft to orbit the Earth 14 times ? (3 marks)
Given : radius of the orbit $r = 6.71\times10^6$ m
radius of the Earth $R_E = 6.37\times10^6$ m

(c) Give **ONE** reason why an aircraft is unable to fly in space like a rocket. (1 mark)

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12.

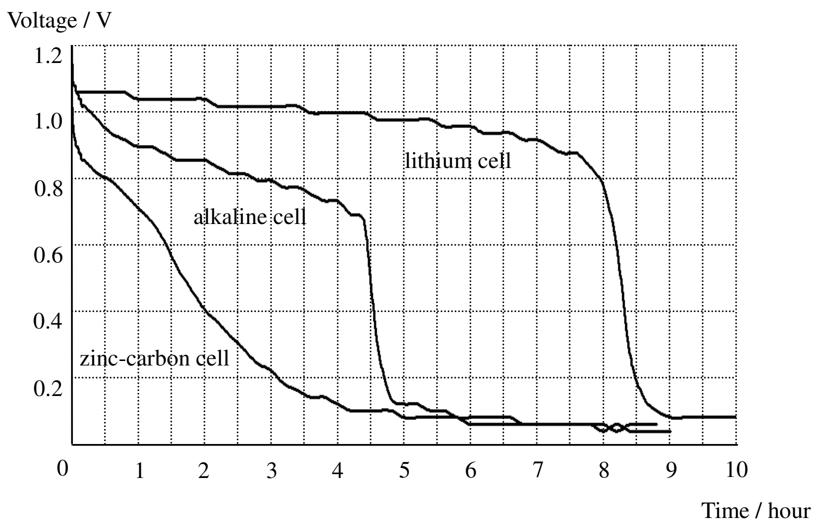
Figure 12.1

Iris uses the apparatus shown in Figure 12.1 to study the lifetime of AA-size cells when used to power a bulb. She connects a cell and a switch to the bulb and uses a voltage sensor to measure the voltage across the bulb.

(a) Draw a circuit diagram to illustrate how the apparatus is connected. Use the symbol (V) to denote the voltage sensor and the data-logger. (2 marks)

(b) Iris conducts the experiment with a zinc-carbon cell, an alkaline cell and a lithium cell separately. Figure 12.2 shows the variation of the voltage across the bulb with time for the cells. The bulb lights up as long as the voltage across it is above 0.6 V.

Figure 12.2



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(b) (i) A salesman claims that the lifetime of a lithium cell for lighting up the bulb is five times that of an alkaline cell. Determine whether the claim is correct or not. (2 marks)

(ii) Table 12.3 shows the prices of the three types of cell.

Type of cells	Price per cell
zinc-carbon	$ 1.5
alkaline	$ 3.8
lithium	$25.0

Table 12.3

Which type of cells is the best buy, in terms of the cost per hour for lighting up the bulb ? Show your calculations. (3 marks)

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13. Josephine conducts an investigation on transformers. Primary and secondary coils are wound on two soft-iron C-cores to form a transformer. She sets up a circuit as shown in Figure 13.1.

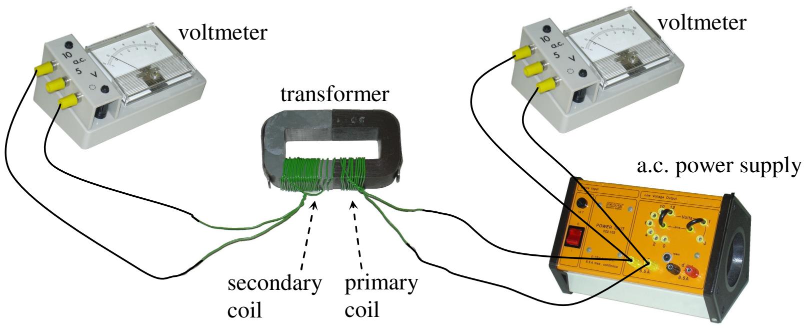

Figure 13.1

*(a) Josephine varies the input voltage V_1 to the transformer and records the corresponding output voltage V_2. The results are shown in Table 13.2. Figure 13.3 shows the graph of V_2 against V_1. Draw a conclusion for this investigation.

V_1 / V	V_2 / V
1.5	2.5
3.0	5.1
4.5	7.6
6.0	10.0

Table 13.2

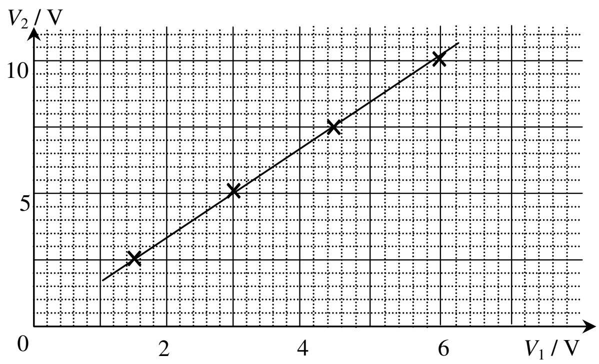

Figure 13.3

(1 mark)

....

....

....

....

*(b) Deduce the value of V_2 that will be produced when V_1 equals 8.0 V. (1 mark)

....

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*(c) Josephine wants to study the relationship between the output voltage and the number of turns in the secondary coil of the transformer. Describe how she can conduct the experiment. (2 marks)

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*(d) Josephine adds a bulb to the circuit as shown in Figure 13.4. Suggest how Josephine can estimate the efficiency of the transformer. State the measurement(s) she must take. Additional apparatus may be used if necessary. (3 marks)

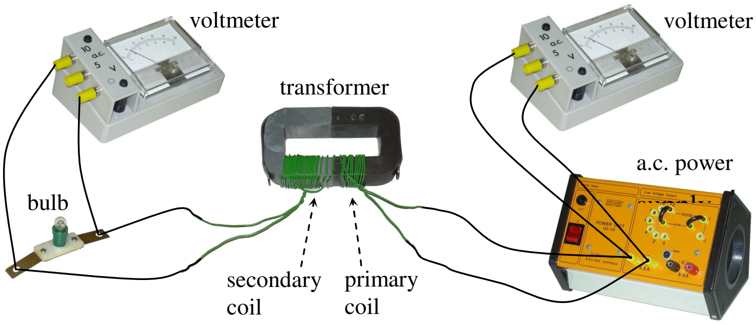

Figure 13.4

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